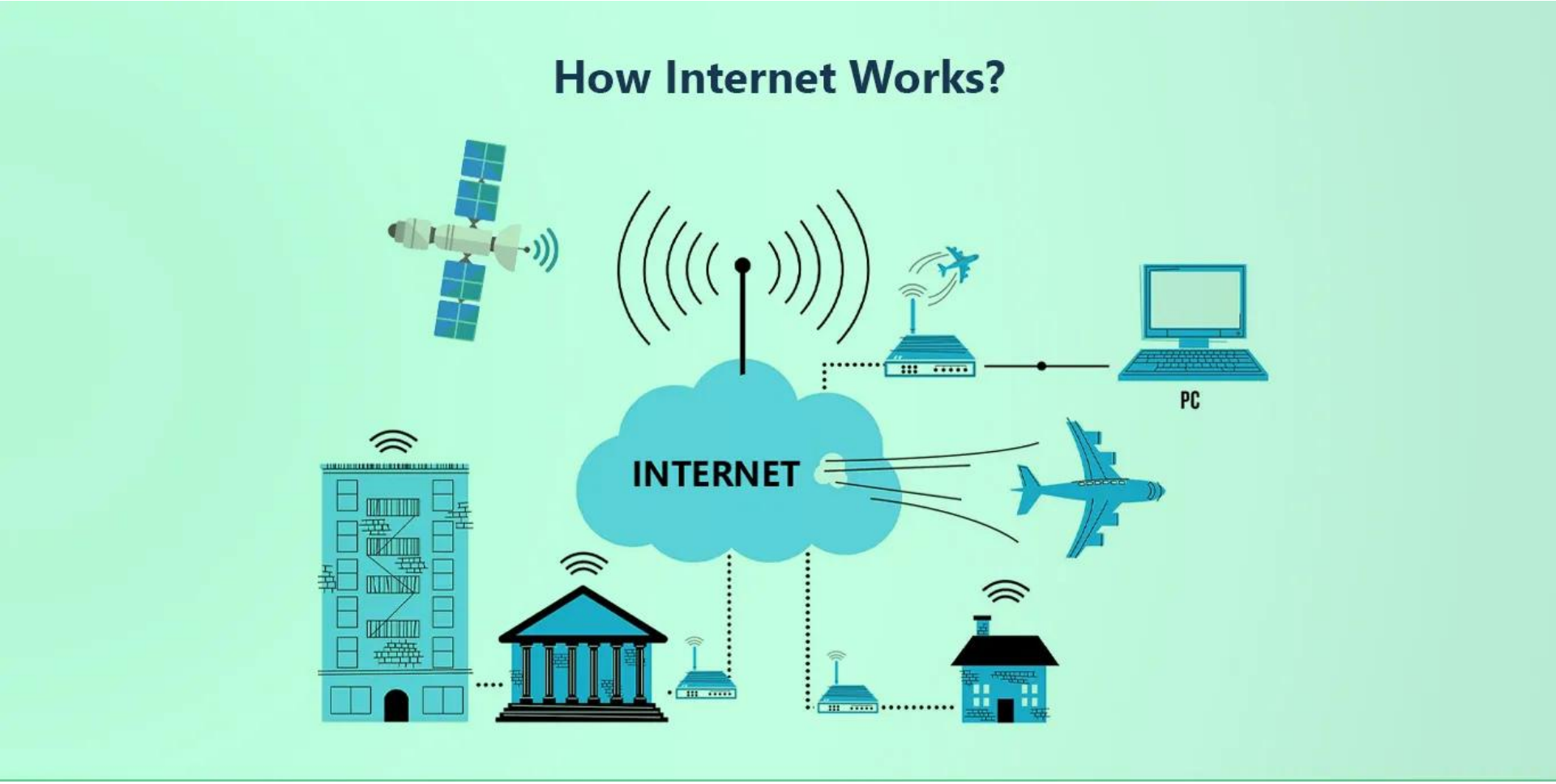
# How the internet works



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**Data Link Layer**

Converts the data packets into network signals for transmission

→ **Data Link Layer** Converts network signal into data packets

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| | |
|---|---|
| **Application Layer** | Web request is received and web server sends the content back |
| **Transport Layer** | Reassemble each packet from full the data request |
| **Internet Layer** | Looks at the IP address of each packets and routes them to the necessary server |
| **Data Link Layer** | Converts network signal into data packets |

# Scenario

You open your browser and type:

www.google.com

and press **Enter**.
What happens behind the scenes?
Your device and the internet use the **TCP/IP model** to make this happen.

# Application Layer (Your Browser)

- You are using Chrome / Safari / Firefox → This is the Application Layer.
- The browser needs the IP Address of www.google.com.
- It asks the ---- server to convert:

**www.google.com → 142.250.185.206 *(example IP for Google)***

**Protocols at this layer:**
✅ ---- / ----
✅ ----

# Transport Layer (TCP)

- The browser now knows the IP.
- The data request is **broken into -----**.
- **TCP** assigns a **port number:**

| Service | Protocol | Port |
|---|---|---|
| Secure Web Browsing | HTTPS | 443 |

•TCP ensures **reliable delivery**:

- Sends packets in order
- Resends any missing packets

# Internet Layer (IP & Routing)

•Each packet receives:

- **Source IP** (your device)
- **Destination IP** (Google server)

•Routers decide the **----** across the Internet.

Example path:

Your PC → Home Router → ISP → Internet Backbone → Google Data Center

# Network / Data Link Layer

This layer converts packets into **signals** over:

•Wi-Fi
•Ethernet cable
•Fiber optic

The packets travel through several devices and networks **very fast**.

# On Google's Server

When packets reach Google's server:

1.Packets are **reassembled** using ----------.
2.Google reads your request (e.g., you searched "weather Dubai").
3.Google prepares a response (webpage, images, suggestions).

# Response Back to You

Google sends the webpage back as packets:

Google → Internet → ISP → Your Router → Your Device

Your browser:

•Reassembles packets
•Renders text, images, and layout
•Displays the Google search page ✅

# Key Summary

You type Google.com
→ DNS finds IP
→ TCP breaks data into packets
→ IP routes packets
→ Server responds
→ You see the webpage

This happens in **milliseconds** ⚡
Thousands of packets per second.

**Question 1:**
Which of the following is an example of a sponsored top level domain?

- .gov
- .com
- .org
- .uk

**Question 2:**
Which of the following layers is responsible for packet acknowledgement

- Internet
- Network
- Transport
- Application

**Question 3:**
Which of the following layers is responsible for converting network signals into data packets

- Physical
- Transport
- Internet
- Data Link